Training and validation curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable checkpoint-evaluation and selection record.

Compute both losses,repair incompatible comparisons and explain development evidence.

Keep one habit: W1–W3/W5 → selected new split comparison → project checkpoint ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Keep the restricted network h = ReLU(x), y = vh, with only v trained. **Training** pairs remain (1, 2) and (2, 4). Start v = 0 and make four full-data updates at rate 0.1. Save states v = 0, 1, 1.5, 1.75 and 1.875.

  At each saved state, evaluate both splits without changing v during evaluation.

- For this deliberately constructed demonstration, validation pairs are (3, 4.5) and (4, 6). Their targets follow a different slope from the **training** pairs.

  This makes the two trends disagree visibly; it is not **evidence** that these few rows represent a real population. Validation labels do not enter the gradient.

- Both columns use ordinary MSE in the same target units, evaluated on their own fixed rows. **Training** MSE is 2.5(v - 2) squared; validation MSE is 12.5(v - 1.5) squared.

  Different **input** values and target rules affect the loss levels. A gap alone does not identify its cause.

Learning rate experiments

> **My remembered idea, example and correction**

Training a small network

> **My remembered idea, example and correction**

Slopes and local change

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Updates	v	Training MSE	Validation MSE
0	0	10	28.125
1	1	2.5	3.125
2	1.5	0.625	0
3	1.75	0.15625	0.78125
4	1.875	0.0390625	1.7578125

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), yvh, savedv 0/1/1.5/1.75/1.875 after0–4 trainingupdates. Trainpairs(1, 2)/(2, 4), MSE 2.5(v–2)²; validation(3, 4.5)/(4, 6), MSE 12.5(v–1.5)². Trainloss10/2.5/.625/.15625/.0390625; validation 28.125/3.125/0/.78125/1.7578125. Lowest validation then earliest exact tie selects index 2, restorev 1.5. Atcheckpoint3 predstrain1.75/3.5, valid 5.25/7. W2 separatetielog[5, 2, 1, 1, 1.2]selects 2.

  SSE 12 over6 meansMSE2, not 12. Known target slopes differ; gapalonecannotdiagnosecomplexity. Separatepictureepochs1–3 train4/2/1, valid 3/2/2.5 select2; changedlastvalid 1.5 select3. Selected validation is development evidence.

- At v = 1, compute both validation **prediction**s, residuals and MSE on (3, 4.5) and (4, 6).

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), yvh, savedv 0/1/1.5/1.75/1.875 after0–4 trainingupdates. Trainpairs(1, 2)/(2, 4), MSE 2.5(v–2)²; validation(3, 4.5)/(4, 6), MSE 12.5(v–1.5)². Trainloss10/2.5/.625/.15625/.0390625; validation 28.125/3.125/0/.78125/1.7578125. Lowest validation then earliest exact tie selects index 2, restorev 1.5. Atcheckpoint3 predstrain1.75/3.5, valid 5.25/7. W2 separatetielog[5, 2, 1, 1, 1.2]selects 2.

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- Observed validation MSE at **checkpoint**s 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which **checkpoint** is selected? What if **training** loss is lowest at **checkpoint** 4?

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), yvh, savedv 0/1/1.5/1.75/1.875 after0–4 trainingupdates. Trainpairs(1, 2)/(2, 4), MSE 2.5(v–2)²; validation(3, 4.5)/(4, 6), MSE 12.5(v–1.5)². Trainloss10/2.5/.625/.15625/.0390625; validation 28.125/3.125/0/.78125/1.7578125. Lowest validation then earliest exact tie selects index 2, restorev 1.5. Atcheckpoint3 predstrain1.75/3.5, valid 5.25/7. W2 separatetielog[5, 2, 1, 1, 1.2]selects 2.

  SSE 12 over6 meansMSE2, not 12. Known target slopes differ; gapalonecannotdiagnosecomplexity. Separatepictureepochs1–3 train4/2/1, valid 3/2/2.5 select2; changedlastvalid 1.5 select3. Selected validation is development evidence.

- A **training** report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says **training** error is four times larger. Repair the comparison.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main hReLU(x), yvh, savedv 0/1/1.5/1.75/1.875 after0–4 trainingupdates. Trainpairs(1, 2)/(2, 4), MSE 2.5(v–2)²; validation(3, 4.5)/(4, 6), MSE 12.5(v–1.5)². Trainloss10/2.5/.625/.15625/.0390625; validation 28.125/3.125/0/.78125/1.7578125. Lowest validation then earliest exact tie selects index 2, restorev 1.5. Atcheckpoint3 predstrain1.75/3.5, valid 5.25/7. W2 separatetielog[5, 2, 1, 1, 1.2]selects 2.

  SSE 12 over6 meansMSE2, not 12. Known target slopes differ; gapalonecannotdiagnosecomplexity. Separatepictureepochs1–3 train4/2/1, valid 3/2/2.5 select2; changedlastvalid 1.5 select3. Selected validation is development evidence.

- **Training** loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development **evidence**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

  Explain the disagreement with selection by **training** loss.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Evaluate new checkpoints independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **training** and **validation curve**s **evidence** sheet to Image-pattern Network Investigation. Show how you can compute comparable **training** and validation losses at saved **checkpoint**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Use the declared rule for **Training** and **validation curve**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Training** and **validation curve**s. Proposed explanation: "The last **checkpoint** is always best". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use h = ReLU(x), y = vh, candidate v values 1, 1.5 and 2. **Training** pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE.

  Explain the disagreement with selection by **training** loss.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Plot **training** and validation loss for **checkpoint**s before opening final cases.

- Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/15

Training and validation curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read training curve

Training loss falls 4 -> 2 -> 1.



2. Read validation curve

Validation loss falls 3 -> 2 then rises to2.5.



3. Choose checkpoint

Select epoch 2, not epoch 3, under the declared validation criterion.

CHANGE ONE THING

Epoch 3 validation loss is instead 1.5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Plot **training** and validation loss for **checkpoint**s before opening final cases.
- Which labelled step matches this detail? Select epoch 2, not epoch 3, under the declared validation criterion.
- 1. Read **training curve**
- 2. Choose **checkpoint**
- 3. Read **validation curve**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Choose **checkpoint** Select epoch 2, not epoch 3, under the declared validation criterion. The **training curve** measures fit to examples used for learning.

The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen. Under the supplied **selection rule**, keep **checkpoint** two.

A later final test must remain outside both learning and **checkpoint** selection.

Changed-case question and optional hint

- Changed-case practice: Epoch 3 validation loss is instead 1.5. Which conclusion is supported by the supplied facts?
- 1. Select epoch 3 under the unchanged criterion.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Select epoch 3 under the unchanged criterion. Its measured validation loss is now below epoch 2′s2; the **training curve** alone never settled selection.

Repair and retain the record

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

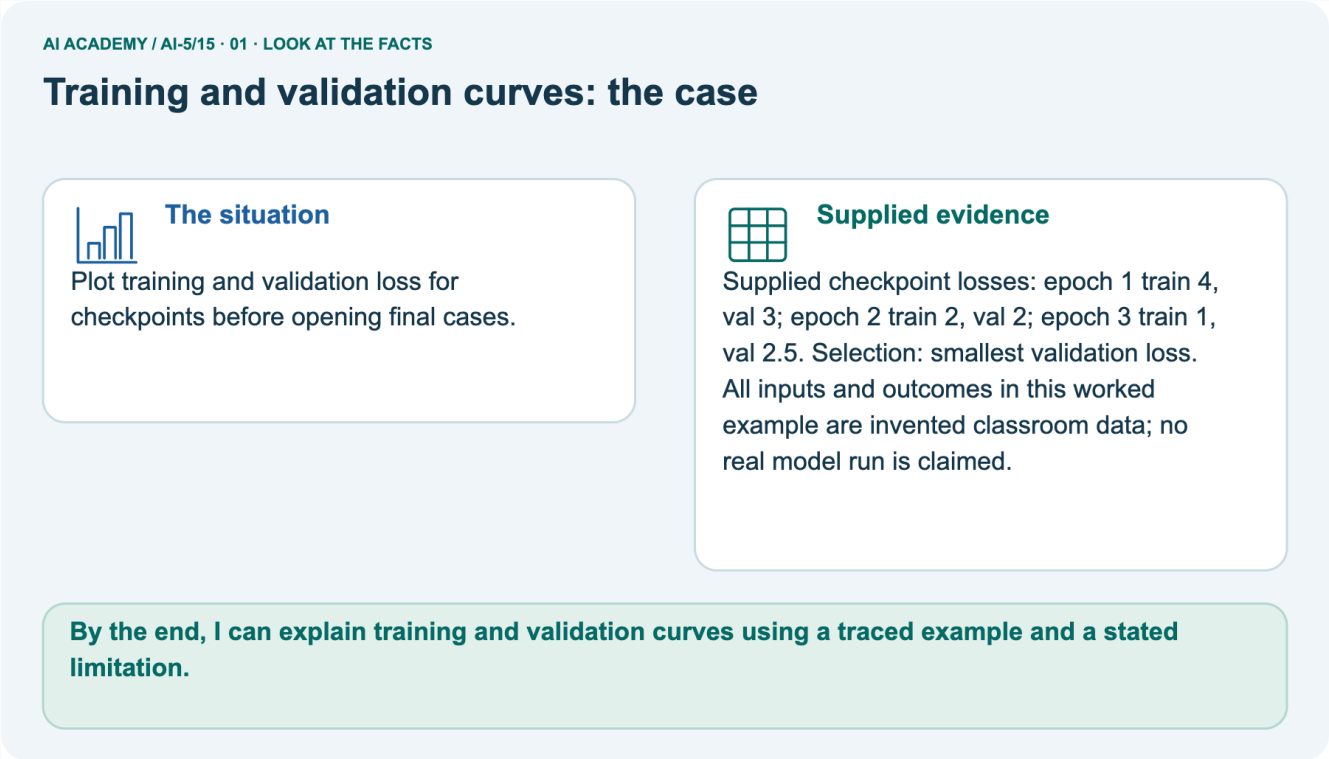
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

Choose the picture that helps explain the current step.

01. Training and validation curves: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

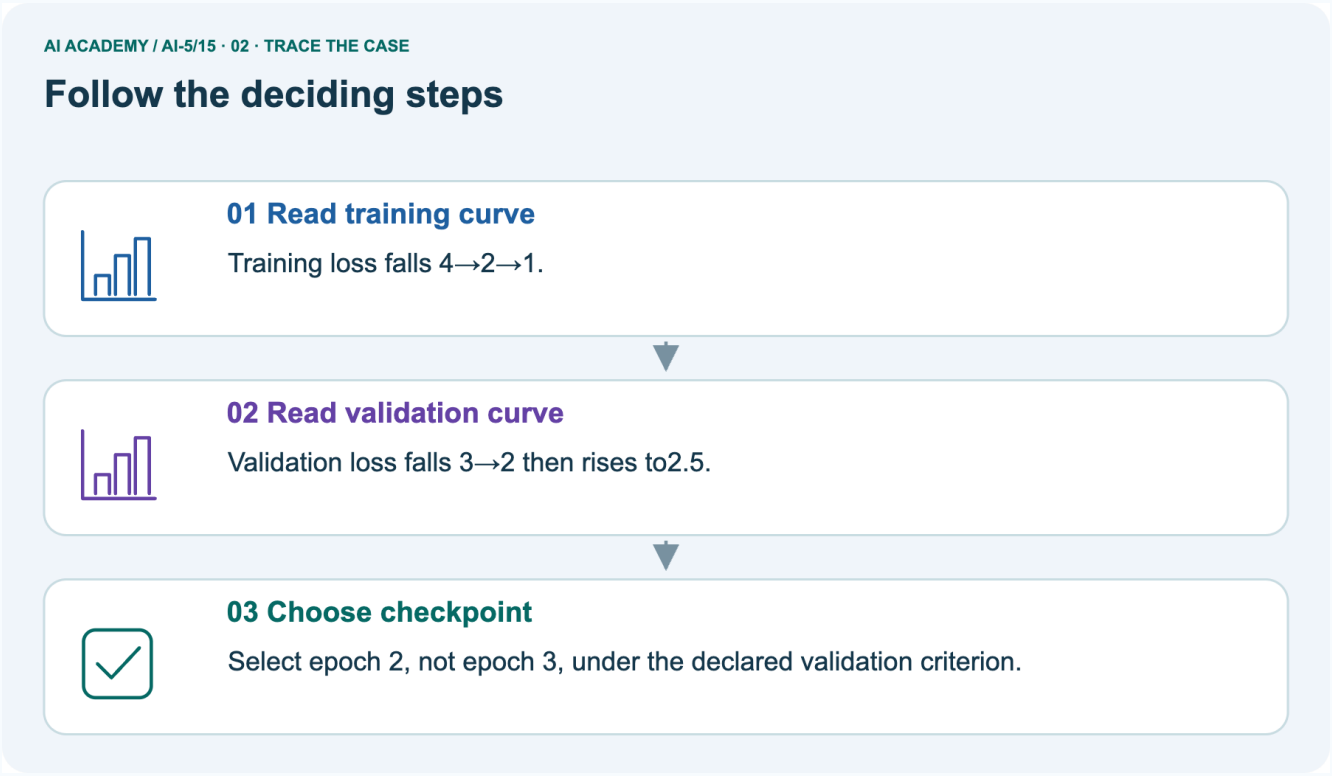
Plot **training** and validation loss for **checkpoint**s before opening final cases. Supplied facts: Supplied **checkpoint** losses: epoch 1 train 4, val 3; epoch 2 train 2, val 2; epoch 3 train 1, val 2.5. Selection: smallest validation loss.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **training curve**: **Training** loss falls 4→2→1. → Read **validation curve**: Validation loss falls 3→2 then rises to2.5. → Choose **checkpoint**: Select epoch 2, not epoch 3, under the declared validation criterion.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Checkpoint losses



Follow the labelled picture with the matching explanation. Open the vector diagram

Checkpoint losses. epoch 1: **training** = 4, validation = 3. epoch 2: **training** = 2, validation = 2. epoch 3: **training** = 1, validation = 2.5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-5/15 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Select epoch 2, not epoch 3, under the declared validation criterion.

Changed condition

Epoch 3 validation loss is instead 1.5.

Worked changed-case result

Select epoch 3 under the unchanged criterion.
Why: Its measured validation loss is now below epoch 2's2; the training curve alone never settled selection.

Follow the labelled picture with the matching explanation. Open the vector diagram

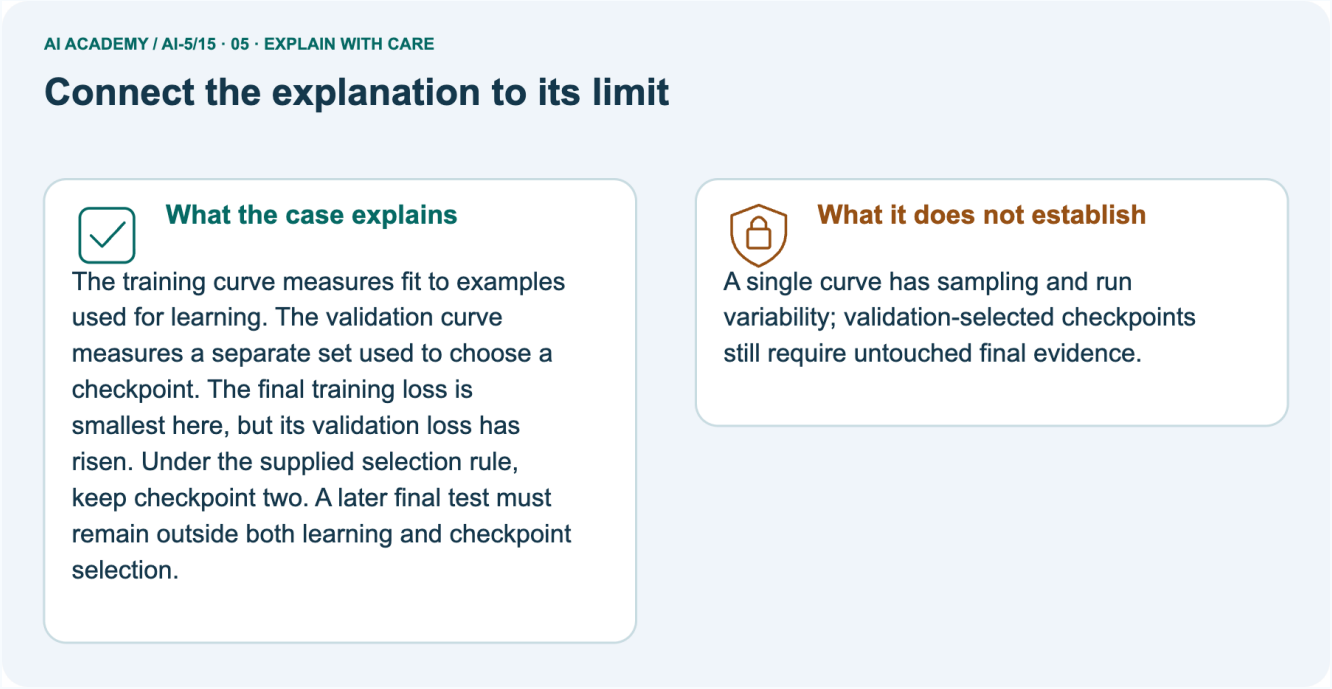
Original case: Select epoch 2, not epoch 3, under the declared validation criterion. Changed condition: Epoch 3 validation loss is instead 1.5. Expected result: Select epoch 3 under the unchanged criterion.

Reason: Its measured validation loss is now below epoch 2's2; the **training curve** alone never settled selection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **training curve** measures fit to examples used for learning. The **validation curve** measures a separate set used to choose a **checkpoint**. The final **training** loss is smallest here, but its validation loss has risen.

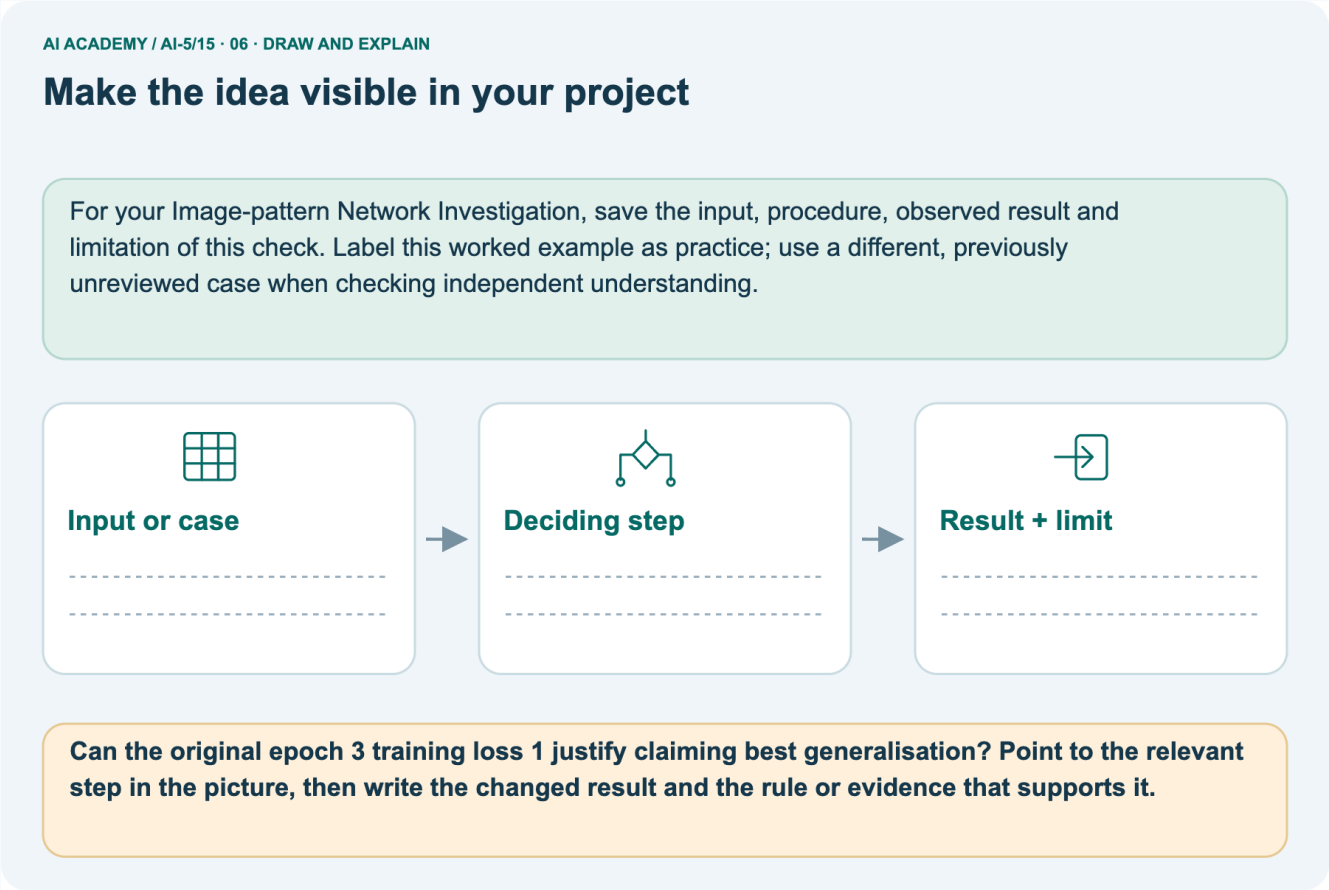
Under the supplied **selection rule**, keep **checkpoint** two. A later final test must remain outside both learning and **checkpoint** selection.

Limit: A single curve has sampling and run variability; validation-selected **checkpoint**s still require untouched final **evidence**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the original epoch 3 **training** loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Training and validation curves: the case

02. Follow the deciding steps

03. Checkpoint losses

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.